CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

BCA-502: Java Programming & Dynamic Webpage Design

Session: December 2016 Examination | **Paper Code:** 18022 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

Q1 (3M) What is Byte-code in JAVA? Explain.

Definition: The intermediate machine instruction set generated by `javac`, executed by the JVM to provide platform independence.

Q2 (3M) What is Garbage Collection in Java?

Definition: Automatic memory reclamation mechanism of the JVM destroying unreferenced heap objects.

Q3 (3M) Differentiate between Class and Interface.

Differences: A class contains full implementation and state; an interface contains method specifications and constants.

Q4 (3M) What is an Applet?

Definition: A small Java program designed to be embedded in an HTML web page and executed inside a web browser.

Q5 (3M) What is Thread in Java?

Definition: A lightweight sub-process representing an independent execution path in a program.

SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

Q6 (7.5M) What do you mean by Constructor? Explain Constructor Overloading.

Constructor Overloading: Defining multiple constructors in a single class with different parameter lists.

Q7 (7.5M) Explain Packages in Java and how to create user-defined packages.

Packages: Directory-based namespaces created with `package` statement.

Q8 (7.5M) Explain Exception Handling in Java with try, catch, and finally.

Handling: Graceful recovery from runtime program faults.

SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

Q9 (15M) Explain Applet Life Cycle in detail with an Applet code to draw shapes.

Applet Shapes Code: Complete applet drawing rectangles, lines, and ovals using `Graphics`.

Q10 (15M) Explain Multithreading and Thread Synchronization with examples.

Multithreading & Synchronization: Producer-Consumer problem and synchronized locks.

Q11 (15M) Explain JDBC Architecture and steps to connect Java to a database.

JDBC: Complete connectivity steps and SQL operations.

Q12 (15M) **Explain Servlets, HttpServlet, and its life cycle methods.**

Servlets: Complete server-side Java web request dispatching.

Q13 (15M) **Explain Client-Server Socket Programming in Java with complete program.**

Socket Programming: `ServerSocket` and `Socket` communication code over TCP.

CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

BCA-502: Java Programming & Dynamic Webpage Design

Session: December 2017 Examination | **Paper Code:** 18022 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

Q1 (3M) **How will you create hyperlinks in a web document? Write the code also.**

HTML Hyperlink:

```
<a href="https://www.ccsuniversity.ac.in" target="_blank">Visit CCSU Official Portal</a>
```

Q2 (3M) **What is a Constructor? List its characteristics.**

Characteristics: Same name as class, no return type (not even void), automatically called on `new`, can be overloaded.

Q3 (3M) **Differentiate between Method Overloading and Method Overriding.**

Overloading: Same method name, different signatures, within same class (compile-time).
Overriding: Same signature, in subclass, overriding parent behavior (runtime).

Q4 (3M) **What is Bytecode in Java?**

Definition: Architecture-neutral intermediate instructions executed by JVM.

Q5 (3M) **What is the purpose of the 'this' keyword?**

Definition: Refers to the current class instance to differentiate shadowed instance fields from parameters.

SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

Q6 (7.5M) **Explain the Life Cycle of an Applet with diagram.**

Applet Lifecycle: Born (`init`) → Running (`start` , `paint`) → Idle (`stop`) → Dead (`destroy`).

Q7 (7.5M) **Explain Packages and Access Protection in Java.**

Access Protection: Detailed comparison of private, default, protected, and public scopes.

Q8 (7.5M) **Explain Interface in Java with code example.**

Interface: Blueprint of a class containing abstract methods and constants.

SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

Q9 (15M) **Explain Exception Handling in Java: try, catch, finally, throw, and throws with custom exception.**

Custom Exception:

```
class InvalidAgeException extends Exception {
    public InvalidAgeException(String msg) { super(msg); }
}
```

Q10 (15M) **Explain Multithreading in Java: Thread class, Runnable interface, and Thread Priorities.**

Multithreading: Full comparison of extending Thread vs implementing Runnable, lifecycle states, and priority scheduling.

Q11 (15M) **What is JDBC? Explain the 4 types of JDBC drivers and steps for database connectivity.**

JDBC Drivers: Type-1, Type-2, Type-3, and Type-4 drivers with architectural diagrams.

Q12 (15M) **Explain Servlets vs JSP. Write a complete HttpServlet program handling GET requests.**

Servlet Architecture: Complete HttpServlet implementation serving dynamic web responses.

Q13 (15M) **Explain Layout Managers in AWT: FlowLayout, BorderLayout, and GridLayout.**

Layout Managers: Visual layout structures and coordinate placements.

CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

BCA-502: Java Programming & Dynamic Webpage Design

Session: December 2018 Examination | **Paper Code:** 18022 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

Q1 (3M) What is a Final Variable? Give an example.

Definition: A variable declared with the `final` keyword becomes a constant whose value cannot be reassigned once initialized:

```
final double PI = 3.14159;
// PI = 3.14; // Compile-time error: cannot assign a value to final variable
```

Q2 (3M) What is Garbage Collection in Java?

Definition: The automatic process of finding and freeing heap memory allocated to objects that are no longer referenced by any active thread.

Q3 (3M) What is an Abstract Method?

Definition: A method declared without a body (implementation) using the `abstract` keyword, terminating with a semicolon. Must be implemented by subclasses.

Q4 (3M) What is a Layout Manager in Java AWT?

Definition: An interface/class that manages the automatic positioning and sizing of GUI components within a container window.

Q5 (3M) What is the purpose of the 'super' keyword?

Definition: References the immediate superclass object to call parent constructors (`super()`) or access hidden parent members.

SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

Q6 (7.5M) What are Packages? How are packages created in Java? Explain and give example.

Packages: Namespaces organizing classes and avoiding naming collisions. Created using the `package` keyword at the top of a source file:

```
package com.ccsu.bca;
public class Student {
    public void display() { System.out.println("BCA Student"); }
}
```

Q7 (7.5M) Explain Interface in Java with an example of Multiple Inheritance.

Explanation: Pure abstraction tool allowing classes to implement multiple interfaces simultaneously, overcoming single class inheritance limits.

Q8 (7.5M) **Explain Applet Life Cycle methods.**

Lifecycle: `init()`, `start()`, `paint(Graphics g)`, `stop()`, and `destroy()`.

SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

Q9 (15M) **Explain Multithreading in Java. Write a program demonstrating Thread synchronization using synchronized methods.**

Thread Synchronization: Coordinates thread access to shared mutable resources to prevent race conditions.

```
class Table {
    synchronized void printTable(int n) {
        for(int i = 1; i ⩽ 5; i++) {
            System.out.println(n * i);
            try { Thread.sleep(300); } catch(Exception e) {}
        }
    }
}
```

Q10 (15M) **What is Exception Handling? Explain Checked vs Unchecked Exceptions with try, catch, finally, and throws.**

Detailed Exception Model: Checked exceptions verified at compile-time (e.g. `IOException`) vs Unchecked exceptions occurring at runtime (e.g. `NullPointerException`).

Q11 (15M) **Explain JDBC Architecture and steps to connect Java to a database. Write a program to insert a record.**

Complete JDBC Program: Register driver, establish connection, execute SQL insert using `Statement`, and close resources.

Q12 (15M) **What is JSP? Explain its life cycle, advantages over Servlets, and scriptlet tags.**

JSP vs Servlets: Translation, compilation, and execution lifecycle; comparison of presentation layer benefits.

Q13 (15M) **Explain Event Handling in Java AWT with Delegation Event Model.**

Delegation Event Model: Event sources generate event objects delivered to registered listener interfaces.

CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

BCA-502: Java Programming & Dynamic Webpage Design

Session: November 2019 Examination | **Paper Code:** 18022 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

Q1 (3M) What is Bytecode and Java Virtual Machine (JVM)?

Explanation: Bytecode is compiled intermediate instructions for the virtual JVM processor, enabling cross-platform portability.

Q2 (3M) List the four fundamental principles of Object-Oriented Programming.

OOP Principles: Encapsulation, Abstraction, Inheritance, and Polymorphism.

Q3 (3M) What is Garbage Collection in Java?

Explanation: Automatic reclamation of heap memory occupied by dereferenced objects.

Q4 (3M) What is Final variable, Final method, and Final class?

Final keyword: Variable becomes a constant, method cannot be overridden, class cannot be inherited (subclassed).

Q5 (3M) Define Constructor in Java.

Definition: A special method called automatically when an object is instantiated, having the same name as the class and no return type.

SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

Q6 (7.5M) Explain Interface in Java with code demonstrating multiple inheritance.

Interface: Contract with method declarations implemented by classes to achieve multiple inheritance.

Q7 (7.5M) Explain Applet Life Cycle and how Applets are embedded into HTML.

Applet in HTML: Embedded using `<applet code="MyApplet.class" width="300" height="200"></applet>`.

Q8 (7.5M) Explain Exception Handling using try, catch, and finally blocks.

Exception Handling: Structured handling preventing runtime crashes and releasing resources safely.

SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

Q9 (15M) What is JSP? Explain advantages of JSP and main components. How is JSP different from Servlet?

JSP vs Servlet: Comprehensive comparison, JSP compilation lifecycle, directives, actions, and scriptlets.

Q10 (15M) Explain Multithreading in Java: Thread Life Cycle, States, and Synchronization.

Thread Life Cycle: New → Runnable → Running → Blocked/Waiting → Terminated.

Q11 (15M) **Explain JDBC Architecture, Statement vs PreparedStatement, and write a program to insert student records.**

JDBC Program: Complete program inserting records using PreparedStatement to prevent SQL injection.

Q12 (15M) **Explain Event Handling in Java AWT/Swing with MouseListener and ActionListener.**

Event Handling: Source → Event Object → Listener interface implementation.

Q13 (15M) **Write HTML and CSS code to design a responsive institutional web page.**

Web Design: Clean HTML markup with external/internal CSS styling, tables, lists, and navigation bar.

CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

BCA-502: Java Programming & Dynamic Webpage Design

Session: December 2020 Examination | **Paper Code:** 18022 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

Q1 (3M) Explain why Java is considered the best language for Internet Applications.

Reasons: Platform independence via bytecode/JVM, built-in security (sandbox security manager), robust memory management, native multithreading, and extensive networking APIs (`java.net`).

Q2 (3M) Discuss various primitive data types supported by Java.

Primitive Types: 8 primitive types: `byte` , `short` , `int` , `long` , `float` , `double` , `char` , `boolean` with explicit fixed sizes across all platforms.

Q3 (3M) What is the purpose of the 'static' keyword in Java?

Explanation: Marks class-level members (variables, methods, blocks) shared by all instances, callable without creating an object of the class (e.g., `ClassName.staticMethod()`).

Q4 (3M) What are Wrapper Classes in Java?

Explanation: Classes that wrap primitive data types into object representations (`Integer` , `Double` , `Boolean`) enabling autoboxing and unboxing for collections.

Q5 (3M) Define Interface in Java.

Definition: A reference type containing pure abstract method declarations and static constants, defining a functional contract for implementing classes.

SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

Q6 (7.5M) Explain Applet Life Cycle and write an Applet to display a greeting message.

Applet Lifecycle: Complete coverage of `init()` , `start()` , `paint()` , `stop()` , and `destroy()` .

Q7 (7.5M) Explain Thread Synchronization and the 'synchronized' keyword in Java.

Synchronization: Prevents thread interference and race conditions when multiple threads access shared resources simultaneously using monitor locks.

Q8 (7.5M) Differentiate between Throw and Throws in Java Exception Handling.

Throw: Explicitly triggers an exception instance inside a method body.
Throws: Declares exceptions thrown by a method in its header signature.

SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

Q9 (15M) What is JDBC? Explain JDBC Architecture and execute a database SELECT query.

JDBC Architecture: Application → JDBC API → DriverManager → JDBC Driver → Database. Complete sample SELECT query program.

Q10 (15M) **Write HTML code for a complete Web Form with Table formatting and Validation.**

HTML Implementation: Form markup with input types, select dropdowns, radio buttons, and table alignments.

Q11 (15M) **Discuss JSP Architecture, Scripting Elements (<%! %>, <% %>, <%= %>), and Directives.**

JSP Scripting Elements:

- **Declaration (`<%! %>`):** Declares class variables and methods.
- **Scriptlet (`<% %>`):** Java code placed inside the `_jspService()` method.
- **Expression (`<%= %>`):** Directly prints evaluated values to the output stream.

Q12 (15M) **Explain Servlets, HttpServletRequest, and HttpServletResponse with complete GET/POST demonstration.**

Servlet Processing: Complete explanation of HTTP request parsing, header inspection, and HTML output streaming.

Q13 (15M) **What are JavaBeans? Explain Bean properties and deployment in JSP using jsp:useBean.**

JavaBeans: Reusable software components adhering to conventions: zero-arg constructor, private fields, and public getters/setters.

CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

BCA-502: Java Programming & Dynamic Webpage Design

Session: December 2021 Examination | **Paper Code:** 18022 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

SECTION A — VERY SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks / CV-3 Pattern)

Q1 (7.5M) **Explain destroy() method, a stage in the life cycle of an Applet.**

Explanation: `destroy()` is the final callback method in the Applet lifecycle, invoked by the browser or AppletViewer before the applet is completely removed from memory and garbage collected. It is called only once after the `stop()` method to release non-memory resources (threads, sockets, database handles).

Q2 (7.5M) **What do you mean by a Package in Java? State the different Access Specifiers available in Java.**

Access Specifiers:

Access Specifier	Same Class	Same Package	Subclass (Diff Pkg)	World (Everywhere)
`private`	Yes	No	No	No
`default` (package-private)	Yes	Yes	No	No
`protected`	Yes	Yes	Yes (via inheritance)	No
`public`	Yes	Yes	Yes	Yes

Q3 (7.5M) **What is Exception? Give different types of exceptions that occur during runtime. Why handle exceptions?**

Explanation: An **Exception** is an abnormal runtime event that disrupts the normal flow of program instructions. Types: Checked (`IOException` , `SQLException`) and Unchecked/Runtime (`ArithmeticException` , `NullPointerException` , `ArrayIndexOutOfBoundsException`). Handled to maintain graceful execution and avoid abrupt program termination.

Q4 (7.5M) **What are the life-cycle methods of a JSP? Differences between JSP Custom Tags and JavaBeans.**

JSP Lifecycle: `jspInit()` → `_jspService(HttpServletRequest, HttpServletResponse)` → `jspDestroy()` .
Differences: JavaBeans encapsulate reusable business data and logic via getter/setter methods. Custom Tags encapsulate presentation logic and HTML formatting within custom XML-style tags.

Q5 (7.5M) **Explain Thread Priority. What are default values? Write a program to define two threads where one prints 1 to 10 and the other prints 10 to 1.**

Thread Priorities: Integer scale from 1 (`MIN_PRIORITY`) to 10 (`MAX_PRIORITY`), with default 5 (`NORM_PRIORITY`).
Program:

```
class ForwardPrinter extends Thread {
    public void run() {
        for(int i = 1; i ⩽ 10; i++) {
            System.out.println("Forward: " + i);
            try { Thread.sleep(200); } catch(Exception e) {}
        }
    }
}

class BackwardPrinter extends Thread {
    public void run() {
        for(int j = 10; j ⩾ 1; j--) {
            System.out.println("Backward: " + j);
            try { Thread.sleep(200); } catch(Exception e) {}
        }
    }
}

public class PriorityThreadDemo {
    public static void main(String[] args) {
        ForwardPrinter t1 = new ForwardPrinter();
        BackwardPrinter t2 = new BackwardPrinter();
        t1.setPriority(Thread.MAX_PRIORITY);
        t2.setPriority(Thread.MIN_PRIORITY);
        t1.start();
        t2.start();
    }
}
```

SECTION B — SHORT ANSWER QUESTIONS (1 × 15 = 15 Marks / CV-3 Pattern)

Q6 (15M) **Explain Client-Server Socket Programming in Java with complete code.**

Socket Programming: `ServerSocket` on server side and `Socket` on client side transmitting data over TCP streams using `DataInputStream` and `DataOutputStream` .

SECTION C — DETAILED ANSWER QUESTIONS (2 × 22.5 = 45 Marks / CV-3 Pattern)

Q7 (22.5M) **Explain JDBC Architecture and write a program to update employee salaries based on department.**

Program:

```
import java.sql.*;

public class UpdateSalaryDemo {
    public static void main(String[] args) {
        try {
            Connection con = DriverManager.getConnection("jdbc:mysql://localhost:3306/db", "root",
            PreparedStatement ps = con.prepareStatement("UPDATE Employee SET Salary = Salary * 1.1
            ps.setString(1, "IT");
            int rows = ps.executeUpdate();
            System.out.println("Salaries updated for " + rows + " employees.");
            con.close();
        } catch(Exception e) { e.printStackTrace(); }
    }
}
```

Q8 (22.5M) **Explain AWT Event Handling with Delegation Event Model. Write a program to handle Button click events.**

Delegation Event Model: An event source generates an event and sends it to registered event listeners implementing listener interfaces (e.g., `ActionListener`).

CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

BCA-502: Java Programming & Dynamic Webpage Design

Session: December 2023 Examination | **Paper Code:** 18022 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

Q1 (3M) How is an Object created in Java?

Object Creation: Uses the `new` operator:

```
ClassName obj = new ClassName();
```

Steps: (1) *Declaration* (associates variable name with object type), (2) *Instantiation* (`new` allocates memory on the Heap), (3) *Initialization* (invokes the constructor).

Q2 (3M) Compare JDK, JRE, and JVM.

Relationship:

- **JVM (Java Virtual Machine):** The runtime engine that interprets and executes bytecode.
- **JRE (Java Runtime Environment):** Contains JVM + Java Standard Class Libraries; sufficient to run Java programs.
- **JDK (Java Development Kit):** Contains JRE + Development Tools (`javac`, `javadoc`, `jar`, `debugger`); needed to write and compile programs.

Q3 (3M) Difference between JDBC and ODBC.

Differences: ODBC is a C/C++ platform-dependent API designed originally for Windows. JDBC is a platform-independent pure Java API tailored for Java applications and cross-platform RDBMS connectivity.

Q4 (3M) Write differences between JSP and Servlet.

Differences: Servlets are Java code with embedded HTML strings (ideal for business logic and controllers). JSP is HTML markup with embedded Java scriptlets (ideal for presentation and views in MVC).

Q5 (3M) What is Method Overloading vs Method Overriding?

Comparison: Overloading occurs in the same class with identical method names but different parameter lists (compile-time polymorphism). Overriding occurs in subclasses with identical method signatures and return types (runtime dynamic dispatch).

SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

Q6 (7.5M) What is an Applet? Describe the complete Life Cycle of an Applet.

Applet Life Cycle: `init()` → `start()` → `paint(Graphics g)` → `stop()` → `destroy()`.

Q7 (7.5M) **Explain Interface in Java. How is multiple inheritance achieved using interfaces?**

Explanation: A class can implement multiple interfaces simultaneously:

```
interface Printable { void print(); }
interface Showable { void show(); }
class Document implements Printable, Showable {
    public void print() { System.out.println("Printing..."); }
    public void show() { System.out.println("Showing..."); }
}
```

Q8 (7.5M) **Explain Exception Handling in Java with try-catch-finally block.**

Handling: Code capable of raising runtime exceptions is placed inside `try`, recovered in `catch`, and resource deallocation performed in `finally`.

SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

Q9 (15M) **What are Packages in Java? How is a user-defined package created and imported? Explain with complete code example.**

Creating & Importing User-Defined Package:

```
// File: mypack/Calculator.java
package mypack;

public class Calculator {
    public int add(int a, int b) { return a + b; }
    public int multiply(int a, int b) { return a * b; }
}

// File: TestPackage.java
import mypack.Calculator;

public class TestPackage {
    public static void main(String[] args) {
        Calculator calc = new Calculator();
        System.out.println("Sum: " + calc.add(10, 20));
        System.out.println("Product: " + calc.multiply(10, 20));
    }
}
```

Q10 (15M) **Define Thread. How to create a thread in Java? Write a program that executes two threads simultaneously.**

Complete Dual Thread Program:

```
class ThreadOne extends Thread {
    public void run() {
        for(int i = 1; i ⩽ 5; i++) {
            System.out.println("Thread 1 executing count: " + i);
            try { Thread.sleep(400); } catch(Exception e) {}
        }
    }
}

class ThreadTwo extends Thread {
    public void run() {
        for(int j = 10; j ⩽ 15; j++) {
            System.out.println("Thread 2 executing count: " + j);
            try { Thread.sleep(400); } catch(Exception e) {}
        }
    }
}

public class DualThreadDemo {
    public static void main(String[] args) {
        ThreadOne t1 = new ThreadOne();
        ThreadTwo t2 = new ThreadTwo();
        t1.start();
        t2.start();
    }
}
```

Q11 (15M) **Explain JDBC Architecture and steps involved in connecting a Java application to a Database. Write complete program.**

Steps: (1) Load Driver, (2) Create Connection, (3) Create Statement, (4) Execute Query, (5) Process ResultSet, (6) Close Connection.

Q12 (15M) **Explain HTML form controls and write code for an Admission Form.**

Admission Form: Includes text fields, dropdown selects, checkboxes, text areas, and submit buttons.

Q13 (15M) **Explain JSP implicit objects and session tracking in Web Applications.**

Implicit Objects: `request`, `response`, `session`, `application`, `out`, `config`, `pageContext`, `page`, `exception`. Session tracking via Cookies, URL Rewriting, and HttpSession.

CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

BCA-502: Java Programming & Dynamic Webpage Design

Session: December 2024 Examination | **Paper Code:** 18022 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

Q1 (3M) Discuss Primitive Data Types in Java.

Java Data Types:

- **Integer Types:** `byte` (1 byte), `short` (2 bytes), `int` (4 bytes), `long` (8 bytes).
- **Floating-Point:** `float` (4 bytes, IEEE 754), `double` (8 bytes).
- **Character:** `char` (2 bytes, 16-bit Unicode character).
- **Boolean:** `boolean` (represents true or false).

Q2 (3M) Explain TCP/IP Server Socket in Java.

Explanation: Java provides the `java.net.ServerSocket` class to implement a TCP server that listens for incoming connection requests on a designated port.

```
ServerSocket server = new ServerSocket(8080);
Socket client = server.accept(); // Blocks until connection arrives
```

Q3 (3M) What are Packages? How do you call a package in Java?

Definition: A **Package** is a namespace mechanism that organizes related classes and interfaces into modular directory structures, preventing naming conflicts and providing access protection.
Calling a Package: Uses the `import` keyword:

```
import java.util.ArrayList; // Imports single class
import java.io.*;           // Imports all classes in package
```

Q4 (3M) Explain the Anchor and Table tags in HTML.

HTML Tags:

- `<a href="url">Link Text</a>` : Creates a hyperlink to navigate between web pages or download files.
- `<table><tr><th>...</th><td>...</td></tr></table>` : Renders structured tabular data using rows (`<tr>`), headers (`<th>`), and data cells (`<td>`).

Q5 (3M) Briefly explain Internet Concepts: URL, HTTP, and Web Server.

Concepts:

- **URL (Uniform Resource Locator):** Global address of a resource on the web (e.g., `https://www.example.com/index.html`).
- **HTTP:** HyperText Transfer Protocol, a stateless client-server protocol running over TCP port 80/443 for web communication.
- **Web Server:** Hardware/software that serves web resources to clients (Apache, Nginx, Tomcat).

SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

Q6 (7.5M) **Explain Applet Life Cycle. Write a program to demonstrate a simple Java applet to display an image. Compare Applets over HTML.**

Applet Image Display Code:

```
import java.applet.*;
import java.awt.*;

/* <applet code="ImageApplet.class" width="400" height="300"></applet> */
public class ImageApplet extends Applet {
    Image img;
    public void init() {
        img = getImage(getDocumentBase(), "college_logo.png");
    }
    public void paint(Graphics g) {
        g.drawImage(img, 20, 20, this);
        g.drawString("CCSU BCA V Semester", 20, 260);
    }
}
```

Comparison over HTML: Applets provide dynamic client-side computation, graphical interactivity, and local state management that static HTML cannot natively provide.

Q7 (7.5M) **Explain TCP/IP Client Socket. Write constructors and methods used to create client sockets.**

Client Socket (java.net.Socket):

- **Constructors:**
  - `Socket(String host, int port)` — Establishes TCP stream connection.
  - `Socket(InetAddress address, int port)`
- **Methods:**
  - `getInputStream()` — Returns InputStream for reading server responses.
  - `getOutputStream()` — Returns OutputStream for sending data.
  - `close()` — Gracefully closes the TCP connection.

Q8 (7.5M) **Explain the concepts of Inheritance in Java and its various types.**

Inheritance: The mechanism where a subclass derives attributes and behaviors from a superclass using `extends`.
Types in Java:

1. **Single Inheritance:** Class B extends Class A.
2. **Multilevel Inheritance:** Class C extends Class B, which extends Class A.
3. **Hierarchical Inheritance:** Class B and Class C both extend Class A.
4. **Multiple Inheritance:** Achieved through **Interfaces** (Java does not support multiple class inheritance to avoid the Diamond Problem).

SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

Q9 (15M) **What is JDBC? Write a complete JDBC program to insert and display records of employees using PreparedStatement.**

Complete PreparedStatement JDBC Program:

```
import java.sql.*;

public class EmployeeJDBC {
    public static void main(String[] args) {
        String url = "jdbc:mysql://localhost:3306/company_db";
        String user = "root";
        String password = "password";

        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection conn = DriverManager.getConnection(url, user, password);

            // 1. Insert record using PreparedStatement
            String insertSql = "INSERT INTO Employee(EmpID, Name, Salary) VALUES(?, ?, ?)";
            PreparedStatement psInsert = conn.prepareStatement(insertSql);
            psInsert.setInt(1, 101);
            psInsert.setString(2, "Rahul Sharma");
            psInsert.setDouble(3, 55000.00);
            int rows = psInsert.executeUpdate();
            System.out.println("Inserted rows: " + rows);

            // 2. Display all employee records
            String selectSql = "SELECT EmpID, Name, Salary FROM Employee";
            Statement stmt = conn.createStatement();
            ResultSet rs = stmt.executeQuery(selectSql);

            System.out.println("\n--- EMPLOYEE RECORDS ---");
            while(rs.next()) {
                System.out.println(rs.getInt("EmpID") + "\t" +
                                   rs.getString("Name") + "\t" +
                                   rs.getDouble("Salary"));
            }
            conn.close();
        } catch(Exception e) {
            e.printStackTrace();
        }
    }
}
```

Q10 (15M) **What are Packages? Explain Java API Packages. List any five methods of the String class with examples.**

String Class Methods:

1. `length()` : Returns character count → `"Hello".length() == 5` .
2. `charAt(int index)` : Returns character at index → `"Hello".charAt(1) == 'e'` .
3. `substring(int begin, int end)` : Extracts substring → `"Java".substring(0, 2) == "Ja"` .
4. `toUpperCase()` : Converts to uppercase → `"bca".toUpperCase() == "BCA"` .
5. `equals(Object another)` : Compares string contents → `"Java".equals("Java") == true` .

Q11 (15M) **Explain Servlets with their Life Cycle. How is it different from the Life Cycle of JSP?**

Comparison:

- **Servlet:** Java class compiled into bytecode; container directly executes `init()`, `service()`, and `destroy()`.
- **JSP:** Text-based document with HTML + embedded Java code; container first translates JSP into a temporary Java Servlet source file, compiles it to bytecode, and invokes `_jspInit()`, `_jspService()`, and `_jspDestroy()`.

Q12 (15M) **Discuss JSP in detail. What are JSP Directives? Explain various types of directives with examples.**

JSP Directives (`<%@ directive attribute="value" %>`):

1. **page Directive:** Controls page-level settings: `<%@ page import="java.util.*, java.sql.*" contentType="text/html" errorPage="error.jsp" %>`
2. **include Directive:** Inserts static files at translation time: `<%@ include file="header.jsp" %>`
3. **taglib Directive:** Imports custom tag libraries (JSTL): `<%@ taglib uri="http://java.sun.com/jsp/jstl/core" prefix="c" %>`

Q13 (15M) **What are Layout Managers? Explain five Layout Managers supported by AWT/Swing.**

Five Layout Managers:

1. **FlowLayout:** Arranges components row-by-row like words in a paragraph.
2. **BorderLayout:** Positions in North, South, East, West, Center regions.
3. **GridLayout:** Rectangular grid of equal-sized cells.
4. **CardLayout:** Stacks components like a deck of cards (one visible at a time).
5. **GridBagLayout:** Most flexible layout manager using `GridBagConstraints` for variable cell spans and weights.

CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

BCA-502: Java Programming & Dynamic Webpage Design

Session: December 2025 Examination | **Paper Code:** 18022 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

Q1 (3M) What is the difference between Process-based and Thread-based multitasking in Java?

Comparison:

Feature	Process-Based Multitasking	Thread-Based Multitasking
Granularity	Heavyweight; each process has its own separate address space.	Lightweight; threads share the same address space in a single process.
Context Switching	High overhead; managed by operating system kernel.	Low overhead; switches quickly within the JVM.
Communication	Requires Inter-Process Communication (IPC, sockets, pipes).	Direct shared memory, wait(), notify(), synchronized blocks.

Q2 (3M) Explain the 'super' keyword in Java with usages.

Usage of 'super': A reference variable used in a subclass to refer directly to its immediate parent class object:

1. **Invoke Parent Constructor:** `super(args);` — must be the first line in subclass constructor.
2. **Access Parent Variables:** `super.variableName` — accesses shadowed parent instance fields.
3. **Call Parent Methods:** `super.methodName()` — invokes overridden parent class methods.

Q3 (3M) What is Bytecode in Java? Why does it enable platform independence?

Explanation: Java source code (`.java`) is compiled by `javac` into an intermediate machine-independent format called **Bytecode** (stored in `.class` files). Any platform equipped with a standard Java Virtual Machine (JVM) interprets and executes this bytecode natively, achieving "Write Once, Run Anywhere" (WORA).

Q4 (3M) Explain Garbage Collection in Java.

Definition: Java features automatic memory management via the Garbage Collector (GC), a background daemon thread that periodically scans the Heap memory and deallocates memory occupied by unreachable (unreferenced) objects, preventing memory leaks without manual pointer freeing.

Q5 (3M) What is the purpose of the 'finally' block in Exception Handling?

Definition: The `finally` block always executes regardless of whether an exception is thrown, caught, or unhandled. Used to perform critical resource clean-up actions such as closing file streams, database connections, and network sockets.

SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

Q6 (7.5M) **Differentiate between Abstract Class and Interface in Java with examples.**

Comparison:

Aspect	Abstract Class	Interface
Methods	Can have abstract and concrete methods with state.	Contains abstract methods (and default/static in Java 8+).
Variables	Can have instance variables (any access modifier).	All fields are implicitly `public static final` constants.
Inheritance	Single inheritance (a class `extends` only 1 abstract class).	Multiple inheritance (a class can `implements` multiple interfaces).

Q7 (7.5M) **Explain Exception Handling mechanism in Java with keywords: try, catch, finally, throw, and throws.**

Keywords & Mechanism:

- `try` : Encloses code that may potentially throw an exception.
- `catch` : Catches and handles specific exception types.
- `finally` : Guaranteed cleanup block executed after try/catch.
- `throw` : Explicitly throws an exception object (e.g., `throw new ArithmeticException("Divide by zero");`).
- `throws` : Declares exceptions in method signatures for callers to handle.

Q8 (7.5M) **Explain the Applet Life Cycle and its core callback methods.**

Lifecycle Methods:

1. `init()` : Called once when the applet is first loaded into the browser/appletviewer to initialize variables.
2. `start()` : Invoked every time the applet becomes visible or browser tab is opened.
3. `paint(Graphics g)` : Renders shapes, text, and images on the screen surface.
4. `stop()` : Invoked when the user leaves the page or minimizes the browser.
5. `destroy()` : Called before the applet is completely reclaimed from memory.

SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

Q9 (15M) **Write an HTML code to create a User Registration Web Form containing: Username, Date of Birth, Address, Gender (Radio buttons), Email-ID, Mobile Number, and Submit/Reset buttons.**

HTML Code:

```
<!DOCTYPE html>
<html>
<head>
    <title>User Registration Form</title>
</head>
<body>
    <h2>Student / User Registration Form</h2>
    <form action="register" method="POST">
        <table cellpadding="6">
            <tr>
                <td>User Name:</td>
                <td><input type="text" name="username" required /></td>
            </tr>
            <tr>
                <td>Date of Birth:</td>
                <td><input type="date" name="dob" required /></td>
            </tr>
            <tr>
                <td>Address:</td>
                <td><textarea name="address" rows="3" cols="30"></textarea></td>
            </tr>
            <tr>
                <td>Gender:</td>
                <td>
                    <input type="radio" name="gender" value="Male" /> Male
                    <input type="radio" name="gender" value="Female" /> Female
                </td>
            </tr>
            <tr>
                <td>Email-ID:</td>
                <td><input type="email" name="email" required /></td>
            </tr>
            <tr>
                <td>Mobile Number:</td>
                <td><input type="tel" name="mobile" pattern="[0-9]{10}" required /></td>
            </tr>
            <tr>
                <td colspan="2" align="center">
                    <input type="submit" value="Register Now" />
                    <input type="reset" value="Clear Form" />
                </td>
            </tr>
        </table>
    </form>
</body>
</html>
```

Q10 (15M) **What is JDBC? Explain the Four Types of JDBC Drivers with architectural diagrams and characteristics.**

Definition: Java Database Connectivity (JDBC) is an industry-standard Java API that enables Java programs to interact with relational databases using SQL queries.

Four Types of JDBC Drivers:

1. **Type-1: JDBC-ODBC Bridge Driver:** Translates JDBC calls to ODBC calls. Requires ODBC driver installation on every client machine. Obsolete and slow.
2. **Type-2: Native-API / Partially Java Driver:** Converts JDBC calls directly into native C/C++ vendor database client libraries (e.g., Oracle OCI). Requires client-side installation.
3. **Type-3: Network Protocol / All-Java Middleware Driver:** Translates JDBC calls into an intermediate server middleware protocol, which then talks to multiple databases. Flexible for 3-tier architectures.
4. **Type-4: Thin Driver / Direct-to-Database Pure Java Driver:** Converts JDBC calls directly into vendor-specific database network protocols via TCP/IP sockets. 100% pure Java, zero client installation, highest performance (Modern standard!).

Q11 (15M) **Explain Servlet Life Cycle and compare it with JSP Life Cycle. Write a complete HttpServlet code example.**

Servlet Life Cycle Methods:

1. `init(ServletConfig config)` : Loads servlet class, creates instance, and initializes parameters (called once).
2. `service(HttpServletRequest req, HttpServletResponse res)` : Dispatches to `doGet()` or `doPost()` for every incoming HTTP request.
3. `destroy()` : Invoked when container unloads the servlet, releasing database and memory resources.

Servlet Implementation Example:

```
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;

public class WelcomeServlet extends HttpServlet {
    public void doGet(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();
        String user = request.getParameter("username");
        out.println("<html><body>");
        out.println("<h2>Welcome, " + user + "!</h2>");
        out.println("<p>Server Response Time: " + new java.util.Date() + "</p>");
        out.println("</body></html>");
    }
}
```

Q12 (15M) **Define Thread in Java. Explain how to create threads using (a) Thread class and (b) Runnable interface. Write a complete multithreaded program.**

Multithreading in Java:

```
// Approach 1: Extending Thread class
class NumberPrinter extends Thread {
    public void run() {
        for(int i = 1; i ⩽ 5; i++) {
            System.out.println("Thread 1: " + i);
            try { Thread.sleep(500); } catch(Exception e) {}
        }
    }
}

// Approach 2: Implementing Runnable interface
class LetterPrinter implements Runnable {
    public void run() {
        for(char c = 'A'; c ⩽ 'E'; c++) {
            System.out.println("Thread 2: " + c);
            try { Thread.sleep(500); } catch(Exception e) {}
        }
    }
}

public class MultiThreadDemo {
    public static void main(String[] args) {
        NumberPrinter t1 = new NumberPrinter();
        Thread t2 = new Thread(new LetterPrinter());
        t1.start(); // Spawns new thread
        t2.start();
    }
}
```

Q13 (15M) **What are Layout Managers in AWT/Swing? Explain BorderLayout, FlowLayout, and GridLayout with examples.**

Layout Managers: Automatically position, size, and rearrange GUI components inside containers regardless of screen resolution or OS window scaling.

- **FlowLayout:** Arranges components sequentially from left to right, wrapping to the next line when space runs out. Default for `Panel` and `Applet`.
- **BorderLayout:** Divides container into 5 regions: `NORTH`, `SOUTH`, `EAST`, `WEST`, and `CENTER`. Default for `Frame` and `JFrame`.
- **GridLayout:** Arranges components into a rectangular grid of equal-sized rows and columns.
